SUB | NIEDERSÄCHSISCHE STAATS- UND UNIVERSITÄTSBIBLIOTHEK GÖTTINGEN

Werk

Titel: Inventiones Mathematicae
Verlag: Springer
Jahr: 1991
Kollektion: Mathematica
Digitalisiert: Niedersächsische Staats- und Universitätsbibliothek Göttingen
Werk Id: PPN356556735_0104
PURL: http://resolver.sub.uni-goettingen.de/purl?PPN356556735_0104

LOG Id: LOG_0022
LOG Titel: On the ...-factor attached to motives.
LOG Typ: article

Übergeordnetes Werk

Werk Id: PPN356556735
PURL: http://resolver.sub.uni-goettingen.de/purl?PPN356556735
OPAC: http://opac.sub.uni-goettingen.de/DB=1/PPN?PPN=356556735

Terms and Conditions

The Goettingen State and University Library provides access to digitized documents strictly for noncommercial educational, research and private purposes and makes no warranty with regard to their use for other purposes. Some of our collections are protected by copyright. Publication and/or broadcast in any form (including electronic) requires prior written permission from the Goettingen State- and University Library.
Each copy of any part of this document must contain there Terms and Conditions. With the usage of the library's online system to access or download a digitized document you accept the Terms and Conditions.
Reproductions of material on the web site may not be made for or donated to other repositories, nor may be further reproduced without written permission from the Goettingen State- and University Library.

For reproduction requests and permissions, please contact us. If citing materials, please give proper attribution of the source.

Contact

Niedersächsische Staats- und Universitätsbibliothek Göttingen
Georg-August-Universität Göttingen
Platz der Göttinger Sieben 1
37073 Göttingen
Germany
Email: gdz@sub.uni-goettingen.de

Invent. math. 104, 245-261 (1991)

Inventiones mathematicae
© Springer-Verlag 1991

On the Γ-factors attached to motives

Christopher Deninger

Mathematisches Institut, Universität Münster, Einsteinstr. 62, W-4400 Münster, Federal Republic of Germany

Oblatum 10-V-1990

0. Introduction

Consider a smooth and projective variety X over a number field k. In [Se2] Serre defines for every place v of k a local Euler factor of the "motive" $H^w(X)$ where $0 \leqq w \leqq 2\dim X$. For the finite places this is essentially a characteristic polynomial of Frobenius acting on l-adic cohomology. For the infinite places v the local factor is given by a formula involving Γ-functions and certain numerical invariants of the Hodge structure of $X_v = X \times_k k_v$ over k_v. There is no obvious analogy to the local factors at the finite places. It is conjectured that the product of the local factors taken over all places of k extends to a meromorphic function on $\mathbb{C}$ with a simple functional equation under the substitution $s \mapsto w+1-s$. All cases where this can be verified use the theory of automorphic L-functions. A proof of the general functional equation is not in sight. As a minimal prerequisite it seems necessary to me to understand the nature of the local factors at infinity in a conceptual way. The present note is adressed to this problem.

Using a simple archimedian analogue B_{ar} of Fontaine's ring B_{DR} we define a new cohomology theory H^*_{ar} called archimedian cohomology for smooth projective varieties over $\mathbb{R}$ or $\mathbb{C}$. The vector spaces H^w_{ar} are infinite dimensional over $\mathbb{R}$ and carry a natural endomorphism Θ. We show that with a suitable definition of an infinite determinant the inverse "characteristic polynomial"

$$\det\left(\frac{s}{2\pi} - \frac{\Theta}{2\pi}\middle| H^w_{ar}(X_v)\right)^{-1}, \quad v|\infty$$

is equal to the local Γ-factor $L_v(H^w(X), s)$ attached to $H^w(X)$ at the infinite place v. Relations of archimedian cohomology to Deligne cohomology are noted in Sect. 5. In the last section we develop the theory of B_{ar} in the spirit of [F1] and [F2]. An archimedian analogue of B_{HT} that differs from B_{ar} only by the absence of the infinitesimal structure has been used before by Fontaine to reformulate classical Hodge theory.

1. Regularized products and infinite determinants

In this section we consider "determinants" of endomorphisms of possibly infinite dimensional real or complex vector spaces. The notion of an infinite determinant which we use seems to have its origins in [R-S]. It is also a standard tool in mathematical physics [H]. We begin with some remarks on regularized products.

(1.1) Consider a finite or infinite sequence (λ_v, α_v) of complex numbers λ_v with a choice of argument α_v. Assume that only finitely many λ_v are zero. The regularized product $\prod \lambda_v = \prod (\lambda_v, \alpha_v)$ is said to exist if the following holds: Let $N \geqq 0$ be such that $\lambda_v \neq 0$ for $v > N$. Then the Dirichlet series $\sum_{v>N} \lambda_v^{-s}$ with $\lambda_v^{-s} = |\lambda_v|^{-s} e^{-is\alpha_v}$ converges absolutely for $\operatorname{Re} s \gg 0$ and has an analytic continuation to a holomorphic function $\zeta_N(s)$ for $\operatorname{Re} s > -\varepsilon$, some $\varepsilon > 0$. Assuming this we define (independent of N)

$$\prod_v (\lambda_v, \alpha_v) = \left(\prod_{v=1}^{N} \lambda_v \right) \exp(-\zeta_N'(0)).$$

For finite sequences we have

$$\prod_{v=1}^{N} (\lambda_v, \alpha_v) = \prod_{v=1}^{N} \lambda_v .$$

Note that the property of being defined and the value of $\prod (\lambda_v, \alpha_v)$ are invariant under a change of finitely many arguments α_v. The following properties for infinite sequences are obvious:

- Existence and value of $\prod_v (\lambda_{\sigma(v)}, \alpha_{\sigma(v)})$ are independent of a chosen bijection σ of the natural numbers.
- If $\mathbb{N} = M \cup M'$ is a partition of the natural numbers such that $\prod_{v \in M} (\lambda_v, \alpha_v)$ and $\prod_{v \in M'} (\lambda_v, \alpha_v)$ exist, then $\prod_v (\lambda_v, \alpha_v)$ exists as well and

$$\prod_v (\lambda_v, \alpha_v) = \left(\prod_{v \in M} (\lambda_v, \alpha_v) \right) \left(\prod_{v \in M'} (\lambda_v, \alpha_v) \right).$$

- For $\lambda \neq 0$, α an argument of λ and any N as above we have

$$\prod_v (\lambda \lambda_v, \alpha + \alpha_v) = \lambda^{N + \zeta_N(0)} \prod_v (\lambda_v, \alpha_v).$$

(1.2) Now we turn our attention to infinite determinants in a simple algebraic setting. Let Θ be an endomorphism of a complex vector space V of countable dimension. We say that $\det_\infty \Theta$ is defined if the following hold:

- V is the direct sum of finite-dimensional Θ-invariant subspaces.
- The eigenvalues λ of Θ occur with finite multiplicities $m(\lambda)$. Thus we may arrange them in a sequence (λ_v) such that each eigenvalue λ occurs exactly $m(\lambda)$ times. We suppose given a choice of argument α_v for each λ_v such that the regularized product $\prod (\lambda_v, \alpha_v)$ exists.

The second condition and the value of $\prod (\lambda_v, \alpha_v)$ are independent of the ordering of the eigenvalues. We set:

$$\det_\infty \Theta = \det_\infty (\Theta | V) = \prod (\lambda_v, \alpha_v).$$

Note the following facts:

- For finite dimensional V we have $\det_\infty \Theta = \det \Theta$.
- Consider a commutative diagram with exact lines

$$\begin{array}{ccccccc} 0 & \to & V' & \to & V & \to & V'' & \to 0 \\ & & \downarrow \Theta' & & \downarrow \Theta & & \downarrow \Theta'' & \\ 0 & \to & V' & \to & V & \to & V'' & \to 0 \end{array}$$

If $\det_\infty \Theta'$ and $\det_\infty \Theta''$ are defined then $\det_\infty \Theta$ is defined as well for the obvious choice of arguments and

$$\det_\infty \Theta = \det_\infty \Theta' \cdot \det_\infty \Theta''.$$

- For $\lambda \neq 0$ and N large enough we have:

$$\det_\infty \lambda\Theta = \lambda^{N+\zeta_N(0)} \det_\infty \Theta.$$

Convention. In the applications Θ will have real eigenvalues with $\lambda_v \to -\infty$ for $v \to \infty$. In order to define the value of the "regularized characteristic polynomial" for some z in $\mathbb{C}$

$$\det_\infty (z\,\mathrm{id}_V - \Theta \mid V)$$

we have to choose arguments $\arg(z-\lambda_v)$ for $v \gg 0$. By assumption $\mathrm{Re}(z-\lambda_v) > 0$ for large v and we take $-\frac{\pi}{2} < \arg(z-\lambda_v) < \frac{\pi}{2}$.

If the infinite determinant is defined for z in a neighborhood U of λ in $\mathbb{C}$ we have

$$\det_\infty (z\,\mathrm{id}_V - \Theta \mid V) = (z-\lambda)^{d_\lambda} f(z) \tag{1.3}$$

where d_λ is the dimension of the λ-eigenspace $V^{\Theta=\lambda}$ of Θ and f is a function on U with $f(\lambda) \neq 0$.

If finally Θ is an endomorphism of a real vector space V we set $V_{\mathbb{C}} = V \otimes \mathbb{C}$ and

$$\det_\infty (z - \Theta \mid V) = \det_\infty (z\,\mathrm{id}_{V_{\mathbb{C}}} - \Theta_{\mathbb{C}} \mid V_{\mathbb{C}})$$

if the right hand side is defined.

2. Some formulas related to the Γ-function

Recall that the Γ-function is analytic in the whole complex plane except for simple poles at the points $z = 0, -1, -2, \ldots$ with residues $\frac{(-1)^n}{n!}$ at $z = -n$. We set

$$\Gamma_{\mathbb{C}}(z) = (2\pi)^{-z}\Gamma(z) \quad \text{and} \quad \Gamma_{\mathbb{R}}(z) = 2^{-\frac{1}{2}} \pi^{-\frac{z}{2}} \Gamma\left(\frac{z}{2}\right).$$

By Legendre's duplication formula or by (2.1) below we have

$$\Gamma_{\mathbb{C}}(z) = \Gamma_{\mathbb{R}}(z)\Gamma_{\mathbb{R}}(z+1).$$

Viewing the inverses of $\Gamma_{\mathbb{C}}$ and $\Gamma_{\mathbb{R}}$ as everywhere holomorphic functions the following equalities hold:

(2.1) Proposition. *For complex z we have:*

$$\left(\frac{1}{\sqrt{2\pi}}\Gamma(z)\right)^{-1} = \prod_{v=0}^{\infty}(v+z)$$

$$\Gamma_{\mathbb{C}}(z)^{-1} = \prod_{v=0}^{\infty}\left(\frac{v}{2\pi}+\frac{z}{2\pi}\right)$$

$$\Gamma_{\mathbb{R}}(z)^{-1} = \prod_{v=0}^{\infty}\left(\frac{v}{\pi}+\frac{z}{2\pi}\right).$$

Here the arguments in the definition of the regularized products are taken in $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ *for* $v \gg 0$.

Proof. Since $\Gamma_{\mathbb{C}}(z+1)=\frac{z}{2\pi}\Gamma_{\mathbb{C}}(z)$, $\Gamma_{\mathbb{R}}(z+2)=\frac{z}{2\pi}\Gamma_{\mathbb{R}}(z)$ we may assume that $\mathrm{Re}\, z>0$. The Hurwitz zeta function

$$\zeta(s,z)=\sum_{v=0}^{\infty}\frac{1}{(v+z)^s}, \qquad \mathrm{Re}\, z>0, \qquad -\frac{\pi}{2}<\arg(v+z)<\frac{\pi}{2}$$

defined originally for $\mathrm{Re}(s)>1$ has a holomorphic continuation to $\mathbb{C}\backslash\{1\}$. It is known that

$$\zeta(0,z)=\tfrac{1}{2}-z$$

and

$$(\partial_s\zeta)(0,z)=\log\Gamma(z)-\tfrac{1}{2}\log 2\pi$$

for a suitable branch of $\log\Gamma(z)$. The second formula seems to be due to Lerch at least for real z. It is proved in [E] (1.10) by a conjunction of Hermite's formula for $\zeta(s,z)$ and Binet's second expression for $\log\Gamma(z)$. For real $z>0$ one can understand the formula conceptually by noting that $(\partial_s\zeta)(0,x)$ is a convex solution of the difference equation

$$f(x+1)-f(x)=\log x$$

and using the equation $\zeta'(0)=-\frac{1}{2}\log 2\pi$.

Thus the regularized products in the proposition exist and are easily seen to have the right values.

3. B_{ar}, part I

In the spirit of Fontaine's approach in [F1], [F2] we will now introduce a very simple archimedian equivalent of B_{DR}. We fix a groundfield K equal to $\mathbb{R}$ or $\mathbb{C}$ in the sequel.

Let $\mathbb{R}(1)$ be the real Tate Hodge-structure over K and consider the infinite real Hodge structures over K

$$\mathrm{Sym}\,\mathbb{R}(1) \quad \text{and} \quad \mathrm{Sym}\,\mathbb{R}(1)\,[\mathbb{R}(-1)].$$

A non-zero element t of the real vector space $\mathbb{R}(1)_{\mathbb{R}}$ underlying $\mathbb{R}(1)$ induces isomorphisms:

$$(\mathrm{Sym}\,\mathbb{R}(1))_{\mathbb{R}} \cong \mathbb{R}[T]$$

and

$$B_{ar,\mathbb{R}} := (\mathrm{Sym}\,\mathbb{R}(1)[\mathbb{R}(-1)])_{\mathbb{R}} \cong \mathbb{R}[T, T^{-1}].$$

The Hodge filtration on

$$B_{ar} = B_{ar,\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C}[T, T^{-1}]$$

is given by

$$\mathrm{Fil}^n B_{ar} = T^{-n}\mathbb{C}[T^{-1}].$$

Complex conjugation c on B_{ar} corresponds to complex conjugation on the coefficients of the Laurent polynomials.

If the groundfield is equal to $\mathbb{R}$ we have an action of the infinite Frobenius F_∞ on $B_{ar,\mathbb{R}}$

$$F_\infty(\sum a_\nu T^\nu) = \sum a_\nu (-1)^\nu T^\nu.$$

The $\mathbb{C}$-linear extension of F_∞ to B_{ar} is denoted by the same symbol.

There is also a canonical infinitesimal structure on B_{ar} namely the action of the (degree-) derivation

$$\Theta_{B_{ar}} = T\frac{d}{dT}.$$

Observe that $\Theta_{B_{ar}}$ is independent of the choice of t which we used to identify B_{ar} with a ring of Laurent polynomials. Define an $\mathbb{R}$-algebra $\mathbb{L} = \mathbb{L}_K$ by:

$$\mathbb{L}_{\mathbb{C}} = (\mathrm{Fil}^0 B_{ar})^{c=\mathrm{id}} \cong \mathbb{R}[T^{-1}]$$

and

$$\mathbb{L}_{\mathbb{R}} = (\mathrm{Fil}^0 B_{ar})^{c=\mathrm{id}, F_\infty = \mathrm{id}} \cong \mathbb{R}[T^{-2}]$$

and endow $\mathbb{L}$ with the derivation $\Theta_{\mathbb{L}} = T\frac{d}{dT}$. Then B_{ar} becomes an $\mathbb{L}$-module and the filtration, the action of complex conjugation c and in case $K = \mathbb{R}$ the action of F_∞ are all compatible with the $\mathbb{L}$-module structure.

We consider the following categories:

$\underline{Hod}_K$ the abelian category of (pure) Hodge structures over K. Objects are finite dimensional bigraded $\mathbb{C}$-vector spaces

$$H = \oplus H^{pq}$$

together with a conjugate linear involution c and in case $K = \mathbb{R}$ a $\mathbb{C}$-linear involution F_∞ such that the following hold: $c(H^{pq}) = H^{qp}$, the inclusion of $H_{\mathbb{R}} = H^{c=\mathrm{id}}$ into H induces an isomorphism $H = H_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ and in case $K = \mathbb{R}$, F_∞ commutes with c and $F_\infty(H^{pq}) = H^{qp}$. Morphisms are $\mathbb{C}$-linear maps commuting with these structures.

$\underline{Der}_K$ the additive category whose objects are free $\mathbb{L} = \mathbb{L}_K$-modules D of finite rank together with an $\mathbb{R}$-linear endomorphism Θ_D such that for all l in $\mathbb{L}$ and d in D we have:

$$\Theta_D(ld) = \Theta_{\mathbb{L}}(l)d + l\Theta_D(d).$$

The last condition is expressed by calling (D, Θ_D) an $(\mathbb{L}, \Theta_{\mathbb{L}})$-module. Morphisms are $\mathbb{L}$-linear maps commuting with the Θ's.

For H in $\underline{Hod}_K$ the tensor product $H \otimes_{\mathbb{C}} B_{ar}$ carries a filtration, a conjugate linear automorphism c and in case $K = \mathbb{R}$ a $\mathbb{C}$-linear involution F_∞. These structures are obtained by taking the tensor product of the corresponding structures on H and B_{ar}. We set:

$$\mathbb{D}(H) = \mathrm{Fil}^0(H \otimes_{\mathbb{C}} B_{ar})^{c=\mathrm{id}} \qquad \text{if } K = \mathbb{C}$$

and

$$\mathbb{D}(H)=\mathrm{Fil}^0(H\otimes_{\mathbb{C}}B_{ar})^{c=\mathrm{id},F_\infty=\mathrm{id}} \quad \text{if } K=\mathbb{R}$$

endowed with the $\mathbb{L}$-module structure and derivation Θ over $\Theta_{\mathbb{L}}$ coming from the ones on B_{ar}.

On the other hand consider D in $\underline{Der}_K$. The tensor product $D\otimes_{\mathbb{L}}B_{ar}$ carries the derivation

$$\Theta=\Theta_D\otimes\mathrm{id}+\mathrm{id}\otimes\Theta_{B_{ar}}$$

over $\Theta_{B_{ar}}$. We set

$$\mathbb{H}(D)=(D\otimes_{\mathbb{L}}B_{ar})^{\Theta=0}=\mathrm{Ker}\,\Theta$$

a $\mathbb{C}$-vector space endowed with the conjugate linear involution $c\hat{=}\mathrm{id}\otimes c$ and in case $K=\mathbb{R}$ with the $\mathbb{C}$-linear involution $F_\infty\hat{=}\mathrm{id}\otimes F_\infty$. The filtration on B_{ar} induces a filtration on $\mathbb{H}(D)$ which is c – and in case $K=\mathbb{R}$ also F_∞-invariant.

The categories $\underline{Hod}_K$, $\underline{Der}_K$ have natural tensor products. By definition the twist $D(n)$ of an object D in $\underline{Der}_K$ has the same underlying $\mathbb{L}$-module as D itself but with the twisted derivation $\Theta_{D(n)}=\Theta_D-n\,\mathrm{id}_D$. Note that

$$D(n)=D\otimes_{\mathbb{L}}\mathbb{L}(n)$$

and that $\mathbb{L}(n)$ is isomorphic to the sub $(\mathbb{L},\Theta)$-module $T^{-n}\mathbb{L}$ of $(B_{ar},\Theta_{B_{ar}})$.

(3.1) Theorem. *The construction $\mathbb{D}$ defines an additive functor commuting with twists*

$$\mathbb{D}:\underline{Hod}_K\to\underline{Der}_K$$

such that for all objects H of $\underline{Hod}_K$ we have:

$$\dim_{\mathbb{C}}H=\mathrm{rk}_{\mathbb{L}}\mathbb{D}(H). \tag{3.2}$$

If $K=\mathbb{C}$ then $\mathbb{D}$ also commutes with $\otimes$.

Proof. First assume that $K=\mathbb{C}$. If $H=H^{pq}\oplus H^{qp}$ and $p<q$ we have:

$$\mathrm{Fil}^0(H\otimes_{\mathbb{C}}B_{ar})=(H^{pq}\otimes_{\mathbb{C}}T^p\mathbb{C}[T^{-1}])\oplus(H^{qp}\otimes_{\mathbb{C}}T^q\mathbb{C}[T^{-1}])$$

and hence

$$\mathbb{D}(H)=[(H^{pq}\otimes_{\mathbb{C}}T^p\mathbb{C}[T^{-1}])\oplus(H^{qp}\otimes_{\mathbb{C}}T^p\mathbb{C}[T^{-1}])]^{c=\mathrm{id}}.$$

Under the isomorphism

$$\mathbb{C}[T^{-1}]=\mathbb{C}\otimes_{\mathbb{R}}\mathbb{L}_{\mathbb{C}}$$

c corresponds to $-\otimes\mathrm{id}$ and hence

$$\mathbb{D}(H)=H_{\mathbb{R}}\otimes T^p\mathbb{L}_{\mathbb{C}}. \tag{3.3}$$

If $H=H^{pp}$ we obtain the same formula and hence (3.2) follows. If H_1, H_2 are Hodge structures of the two types discussed above with $p=p_1$ resp. $p=p_2$ we have

$$\begin{aligned}\mathbb{D}(H_1\otimes H_2)&=(H_1\otimes_{\mathbb{C}}H_2)_{\mathbb{R}}\otimes_{\mathbb{R}}T^{p_1+p_2}\mathbb{L}_{\mathbb{C}}\\&\cong(H_{1\,\mathbb{R}}\otimes_{\mathbb{R}}H_{2\,\mathbb{R}})\otimes_{\mathbb{R}}(T^{p_1}\mathbb{L}_{\mathbb{C}}\otimes_{\mathbb{L}_{\mathbb{C}}}T^{p_2}\mathbb{L}_{\mathbb{C}})\\&\cong\mathbb{D}(H_1)\otimes_{\mathbb{L}_{\mathbb{C}}}\mathbb{D}(H_2).\end{aligned}$$

Hence $\mathbb{D}$ commutes with tensor products and also with twists because of the canonical $(\mathbb{L}, \Theta)$-isomorphism

$$\mathbb{D}(\mathbb{R}(1)) = \mathbb{R}(1)_{\mathbb{R}} \otimes T^{-1}\mathbb{L}_{\mathbb{C}} \cong \mathbb{L}_{\mathbb{C}}(1)$$

induced by mapping $t \otimes T^{-1}$ to 1.

Now we assume that $K = \mathbb{R}$. If $H = H^{pq} \oplus H^{qp}$ and $p < q$ or $H = H^{pp}$ we have:

$$\mathbb{D}(H) = (H_{\mathbb{R}} \otimes_{\mathbb{R}} T^p \mathbb{L}_{\mathbb{C}})^{F_\infty = \mathrm{id}}.$$

Using the $(\mathbb{L}_{\mathbb{R}}, \Theta_{\mathbb{L}_{\mathbb{R}}})$-isomorphism

$$T^p \mathbb{L}_{\mathbb{C}} = T^p \mathbb{L}_{\mathbb{R}} \oplus T^{p-1} \mathbb{L}_{\mathbb{R}}$$

we find

$$\mathbb{D}(H) = (H_{\mathbb{R}}^{F_\infty = (-1)^p} \otimes_{\mathbb{R}} T^p \mathbb{L}_{\mathbb{R}}) \oplus (H_{\mathbb{R}}^{F_\infty = (-1)^{p-1}} \otimes_{\mathbb{R}} T^{p-1} \mathbb{L}_{\mathbb{R}}). \tag{3.4}$$

Again this implies (3.2).

Compatibility with twists results from a simple computation. Finally since F_∞ acts by multiplication with -1 on $\mathbb{R}(1)_{\mathbb{R}}$ we have canonically

$$\mathbb{D}(\mathbb{R}(1)) = \mathbb{R}(1)_{\mathbb{R}} \otimes_{\mathbb{R}} T^{-1} \mathbb{L}_{\mathbb{R}} \cong \mathbb{L}_{\mathbb{R}}(1).$$

Warning: For $K = \mathbb{R}$ the functor $\mathbb{D}$ does not in general commute with $\otimes$.

Remark: We have $\mathbb{H}(\mathbb{D}(H)) \cong H$ naturally in the category of complex vector spaces (with involution F_∞ if $K = \mathbb{R}$). However it is not possible to recover the Hodge structure on H from $\mathbb{D}(H)$ functorially. If H is pure of weight w then at least the isomorphism class of H is determined by $\mathbb{D}(H)$. A more suitable setting for the theory of B_{ar} will be given in Sect. 6.

For the moment let us just note the following compatibility. Consider the following functors:

$R_{\mathbb{C}/\mathbb{R}}$: $\underline{Hod}_{\mathbb{C}} \to \underline{Hod}_{\mathbb{R}}$ which maps H to $H \otimes H$ with F_∞ given by $F_\infty(h_1, h^2) = (h_2, h_1)$.

$R_{\mathbb{C}/\mathbb{R}}$: $\underline{Der}_{\mathbb{C}} \to \underline{Der}_{\mathbb{R}}$ which maps an $(\mathbb{L}_{\mathbb{C}}, \Theta)$-module D to itself but viewed as an $\mathbb{L}_{\mathbb{R}}$-module.

(3.5) Proposition. *We have* $\mathbb{D} \circ R_{\mathbb{C}/\mathbb{R}} \cong R_{\mathbb{C}/\mathbb{R}} \circ \mathbb{D}$ *i.e. the diagram*

$$\begin{array}{ccc} \underline{Hod}_{\mathbb{C}} & \xrightarrow{R_{\mathbb{C}/\mathbb{R}}} & \underline{Hod}_{\mathbb{R}} \\ \downarrow \mathbb{D} & & \downarrow \mathbb{D} \\ \underline{Der}_{\mathbb{C}} & \xrightarrow{R_{\mathbb{C}/\mathbb{R}}} & \underline{Der}_{\mathbb{R}} \end{array}$$

commutes up to isomorphism of functors.

Proof. A natural equivalence between $\mathbb{D} \circ R_{\mathbb{C}/\mathbb{R}}$ and $R_{\mathbb{C}/\mathbb{R}} \circ \mathbb{D}$ is given a follows. On objects $H = H^{pq} \oplus H^{qp}$ or $H = H^{pp}$ the map between

$$\begin{aligned} &\mathbb{D}(R_{\mathbb{C}/\mathbb{R}} H) \\ &= ((H_{\mathbb{R}} \oplus H_{\mathbb{R}})^{F_\infty = (-1)^p} \otimes_{\mathbb{R}} T^p \mathbb{L}_{\mathbb{R}}) \oplus ((H_{\mathbb{R}} \oplus H_{\mathbb{R}})^{F_\infty = (-1)^{p-1}} \otimes_{\mathbb{R}} T^{p-1} \mathbb{L}_{\mathbb{R}}) \end{aligned}$$

and

$$R_{\mathbb{C}/\mathbb{R}}\mathbb{D}(H)=(H_{\mathbb{R}}\otimes_{\mathbb{R}}T^{p}\mathbb{L}_{\mathbb{R}})\oplus(H_{\mathbb{R}}\otimes_{\mathbb{R}}T^{p-1}\mathbb{L}_{\mathbb{R}})$$

is induced by the projection onto the first factor:

$$pr_1 : H_{\mathbb{R}}\oplus H_{\mathbb{R}}\to H_{\mathbb{R}}.$$

4. Archimedian cohomology and the Γ-factors

For a real Hodge structure H over $K=\mathbb{C}$ or $\mathbb{R}$ we set following Serre [Se2]:

$$L_{\mathbb{C}}(H,s)=\prod_{p,q}\Gamma_{\mathbb{C}}(s-\min(p,q))^{h^{pq}}$$

and

$$L_{\mathbb{R}}(H,s)=\prod_{p<q}\Gamma_{\mathbb{C}}(s-p)^{h^{pq}}\prod_{p}\Gamma_{\mathbb{R}}(s-p)^{h^{p,+}}\Gamma_{\mathbb{R}}(s-p+1)^{h^{p,-}}$$

where $h^{pq}=\dim_{\mathbb{C}}H^{pq}$ and $h^{p,\pm}$ is the dimension of the $\pm(-1)^p$ eigenspace of F_∞ on H^{pp}. Our main result is the following:

(4.1) Theorem. *For complex s the regularized determinant*

$$\det_\infty\left(\frac{s}{2\pi}-\frac{\Theta}{2\pi}\middle|\mathbb{D}(H)\right)$$

exists in the sense and with the convention of (1.2) *and we have:*

$$\det_\infty\left(\frac{s}{2\pi}-\frac{\Theta}{2\pi}\middle|\mathbb{D}(H)\right)^{-1}=L_K(H,s). \tag{4.2}$$

Proof. We begin with the case $K=\mathbb{R}$. Since both sides of the equation are multiplicative on finite direct sums in $\underline{Hod}_{\mathbb{R}}$ we may assume that

$$H=H^{pp}\quad\text{or}\quad H=H^{pq}\oplus H^{qp}\quad\text{with}\quad p<q.$$

According to (3.4) we have:

$$\mathbb{D}(H)\cong(H_{\mathbb{R}}^{F_\infty=(-1)^p}\otimes_{\mathbb{R}}\mathbb{L}_{\mathbb{R}}(-p))\oplus(H_{\mathbb{R}}^{F_\infty=(-1)^{p-1}}\otimes_{\mathbb{R}}\mathbb{L}_{\mathbb{R}}(-p+1)).$$

The conditions of (1.2) are satisfied for the endomorphism $\frac{s}{2\pi}-\frac{\Theta}{2\pi}$ of $\mathbb{D}(H)_{\mathbb{C}}$ and using (2.1) we find

$$\det_\infty\left(\frac{s}{2\pi}-\frac{\Theta}{2\pi}\middle|\mathbb{D}(H)\right)=\left[\prod_{\nu=0}^{\infty}\left(\frac{s}{2\pi}-\frac{p-2\nu}{2\pi}\right)\right]^{h^+}\left[\prod_{\nu=0}^{\infty}\left(\frac{s}{2\pi}-\frac{p-1-2\nu}{2\pi}\right)\right]^{h^-}$$

$$=\Gamma_{\mathbb{R}}(s-p)^{-h^+}\Gamma_{\mathbb{R}}(s-p+1)^{-h^-}$$

where we have set:

$$h^{\pm}=\dim_{\mathbb{R}}H_{\mathbb{R}}^{F_\infty=\pm(-1)^p}.$$

If $H=H^{pp}$ then $h^{\pm}=h^{p,\pm}$ and (4.2) follows. If $H=H^{pq}\oplus H^{qp}$ where $p<q$ we have $h^+=h^-=h^{pq}$ and thus we get

$$\det_\infty\left(\frac{s}{2\pi}-\frac{\Theta}{2\pi}\middle|\mathbb{D}(H)\right)^{-1}=\Gamma_{\mathbb{C}}(s-p)^{h^{pq}}$$

employing the formula

$$\Gamma_{\mathbb{C}}(s)=\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1)\ .$$

The case $K=\mathbb{C}$ can be treated similarly. It is also a consequence of the case $K=\mathbb{R}$ because for H in $\underline{Hod}_{\mathbb{C}}$ we have:

$$\begin{aligned}\det_\infty\left(\frac{s}{2\pi}-\frac{\Theta}{2\pi}\middle|\mathbb{D}(H)\right) &= \det_\infty\left(\frac{s}{2\pi}-\frac{\Theta}{2\pi}\middle|R_{\mathbb{C}/\mathbb{R}}\mathbb{D}(H)\right)\\ &\overset{(3.5)}{=} \det_\infty\left(\frac{s}{2\pi}-\frac{\Theta}{2\pi}\middle|\mathbb{D}(R_{\mathbb{C}/\mathbb{R}}H)\right)\\ &= L_{\mathbb{R}}(R_{\mathbb{C}/\mathbb{R}}H,s)^{-1}\\ &= L_{\mathbb{C}}(H,s)^{-1}\ .\end{aligned}$$

For a smooth projective variety X over $K=\mathbb{R}$ or $\mathbb{C}$ we define "archimedian" cohomology groups by

$$H^w_{ar}(X)=\mathbb{D}(H^w_B(X_{\mathbb{C}},\mathbb{C}))$$

where the Betti cohomology groups

$$H^w_B(X_{\mathbb{C}},\mathbb{C})=H^w_{\mathrm{sing}}(X(\mathbb{C}),\mathbb{C})$$

are viewed as real Hodge structures over K. It is clear that H^w_{ar} factorizes over the category of absolute Hodge motives M over K. Moreover $H^w_{ar}(M)$ is a free $(\mathbb{L},\Theta)$-module whose rank equals the w-th Betti number of M by (3.2). According to the properties of $\mathbb{D}$ noted in (3.1) the theory H^*_{ar} has a Künneth formula for $K=\mathbb{C}$ but not for $K=\mathbb{R}$. In both cases however H^*_{ar} commutes with twists.

If M is an absolute Hodge motive over a number field k we set for every archimedian place v of k:

$$L_v(H^w(M),s)=L_{k_v}(H^w_B(M),s)$$

and we can consider the archimedian cohomology groups $H^w_{ar}(M_v)$ of $M_v=M\otimes_k k_v$. According to (4.1) we have

$$L_v(H^w(M),s)=\det_\infty\left(\frac{s}{2\pi}-\frac{\Theta}{2\pi}\middle|H^w_{ar}(M_v)\right)^{-1}\ . \tag{4.3}$$

Apart from the obvious analogy between the formulas for L_v at finite and infinite places v there is also a crucial difference: For $v|\infty$, L_v is a "characteristic polynomial" in s whereas for $v\nmid\infty$ *the* L_v are "characteristic polynomials" in the various transcendental functions Nv^{-s}. This suggests to look for an infinite dimensional real cohomology theory with endomorphism $\Theta_v=$"$\log_v Fr_v$" also for the *finite* primes of k such that L_v is given by a regularized "characteristic polynomial" of Θ_v in s itself and not in Nv^{-s}. Note in this direction that even the simplest Euler factor of all $(1-p^{-s})^{-1}$ is essentially given by an infinite product:

$$p^{\frac{s}{2}}-p^{-\frac{s}{2}}=2i\sin\pi\left(\frac{s}{2\pi i}\log p\right)=s\log p\prod_{\nu=1}^{\infty}\left(1+\frac{s^2}{4\pi^2\nu^2}\log^2 p\right)$$

or by regularized infinite products

$$p^{\frac{s}{2}} - p^{-\frac{s}{2}} = i \prod_{\nu=0}^{\infty} \left(\nu + \frac{s}{2\pi i} \log p\right) \prod_{\nu=1}^{\infty} \left(\nu - \frac{s}{2\pi i} \log p\right).$$

We end these speculations by pointing out that there should be a natural geometric definition of $(H^w_{ar}(M), \Theta)$ independent of Hodge theory such that our definition via B_{ar} becomes a theorem.

5. Relations of H_{ar} with Deligne cohomology

Consider a real Hodge structure H over $K = \mathbb{R}$ or $\mathbb{C}$. According to (4.1) and the earlier remark (1.3) the pole order of $L_K(H, s)$ at the integer $s = m$ is given by:

$$\operatorname{ord}_{s=m} L_K(H, s) = -\dim_{\mathbb{R}} \mathbb{D}(H)^{\Theta = m}.$$

In particular if M is a motive over the number field k we find

$$\operatorname{ord}_{s=m} L_v(H^w(M), s) = -\dim_{\mathbb{R}} H^w_{ar}(M_v)^{\Theta = m}$$

for every infinite place v of k. On the other hand the pole order of $L_v(H^w(M), s)$ is related to Deligne cohomology. The following observation is due to Beilinson [B] (3.1), (3.3) see also [Sch] Sect. 2:

(5.1) Proposition. *Assume $m \leqq \frac{w}{2}$ and set $n = w + 1 - m$. Then*

$$\operatorname{ord}_{s=m} L_v(H^w(M), s) = -\dim_{\mathbb{R}} H^{w+1}_{\mathcal{D}}(M_v, \mathbb{R}(n)).$$

Observe that for $m \leqq \frac{w}{2}$ Deligne cohomology factorizes over the category of absolute Hodge motives (use (5.2) below).

The proposition will be explained algebraically by isomorphisms between

$$H^w_{ar}(M_v)^{\Theta = m} \quad \text{and (a twist of)} \quad H^{w+1}_{\mathcal{D}}(M_v, \mathbb{R}(n))$$

for $m \leqq \frac{w}{2}$. The situation is not completely satisfactory however since the isomorphisms depend on choices. For the rest of the section let M be a motive over K. Since $m \leqq \frac{w}{2}$ there is an exact sequence [J] (4.9):

$$0 \to H^w_B(M, \mathbb{R}(n)) \to H^w_B(M, \mathbb{C})/\mathrm{Fil}^n \to H^{w+1}_{\mathcal{D}}(M, \mathbb{R}(n)) \to 0. \tag{5.2}$$

To calculate $H^w_{ar}(M)^{\Theta = m}$ we first assume that $K = \mathbb{C}$:

$$\begin{aligned} H^w_{ar}(M)^{\Theta = m} &= \mathrm{Fil}^0 (H^w_B(M, \mathbb{C}) \otimes_{\mathbb{C}} B_{ar})^{c = \mathrm{id}, \Theta = m} \\ &= \left(\sum_{\nu} \mathrm{Fil}^{\nu} H^w_B(M, \mathbb{C}) \otimes_{\mathbb{C}} \bigoplus_{k \leqq \nu} \mathbb{R}(k) \right)^{c = \mathrm{id}, \Theta = m} \\ &= (\mathrm{Fil}^m H^w_B(M, \mathbb{C}) \otimes_{\mathbb{C}} \mathbb{R}(m))^{c = \mathrm{id}}. \end{aligned}$$

Hence we also have:

$$\begin{aligned} H^w_{ar}(M)^{\Theta = m} &= (\overline{\mathrm{Fil}^m} H^w_B(M, \mathbb{C}) \otimes_{\mathbb{C}} \mathbb{R}(m))^{c = \mathrm{id}} \quad \text{for} \quad K = \mathbb{C} \\ &= (\overline{\mathrm{Fil}^m} H^w_B(M, \mathbb{C}))^{c = (-1)^m} \end{aligned}$$

and similarly

$$H^w_{ar}(M)^{\Theta=m} = (\overline{\mathrm{Fil}^m} H^w_B(M_{\mathbb{C}}, \mathbb{C}) \otimes_{\mathbb{C}} \mathbb{R}(m))^{c=\mathrm{id}, F_\infty=\mathrm{id}} \quad \text{for} \quad K=\mathbb{R}$$

$$= (\overline{\mathrm{Fil}^m} H^w_B(M_{\mathbb{C}}, \mathbb{C}))^{c=(-1)^m, F_\infty=\mathrm{id}}.$$

The Hodge filtration is w-opposed to its complex conjugate i.e.

$$\mathrm{Fil}^n \oplus \overline{\mathrm{Fil}^m} = H^w_B(M_{\mathbb{C}}, \mathbb{C}). \tag{5.3}$$

Now assume that $K=\mathbb{C}$ and consider the map:

$$H^w_{ar}(M)^{\Theta=m} \xrightarrow{\tau_w} H^{w+1}_{\mathscr{D}}(M, \mathbb{R}(n))$$

which is induced by mapping ξ in $\overline{\mathrm{Fil}^m} H^w_B(M, \mathbb{C})$ with $c(\xi)=(-1)^m\xi$ to $(2\pi i)^w \xi$ in $H^w_B(M, \mathbb{C})$.

We have isomorphisms

$$\overline{\mathrm{Fil}^m} \cap H^w_B(M, \mathbb{R}(n+1)) \xrightarrow{\sim} \frac{H^w_B(M, \mathbb{R}(n+1))}{\pi_{n+1}(\mathrm{Fil}^n)} \xrightarrow{\sim} \frac{H^w_B(M, \mathbb{C})}{H^w_B(M, \mathbb{R}(n)) + \mathrm{Fil}^n}$$

and hence τ_w is an isomorphism as well. Here $\pi_a(z)=\frac{1}{2}(z+(-1)^a\bar{z})$.

Now assume that $K=\mathbb{R}$. Then τ_w applied to the cohomologies of $M_{\mathbb{C}}$ induces an isomorphism

$$H^w_{ar}(M)^{\Theta=m} \xrightarrow{\tau_w}_{\sim} H^{w+1}_{\mathscr{D}}(M, \mathbb{R}(n)) = H^{w+1}_{\mathscr{D}}(M_{\mathbb{C}}, \mathbb{R}(n))^{F_\infty=\mathrm{id}}$$

if w is *even*. For *odd* w we only get an isomorphism

$$H^w_{ar}(M)^{\Theta=m} \xrightarrow{\tau_w}_{\sim} H^{w+1}_{\mathscr{D}}(M, \mathbb{R}(n))^- := H^{w+1}_{\mathscr{D}}(M_{\mathbb{C}}, \mathbb{R}(n))^{F_\infty=-\mathrm{id}}.$$

In this case however

$$H^{w+1}_{\mathscr{D}}(M, \mathbb{R}(n))^- \quad \text{and} \quad H^{w+1}_{\mathscr{D}}(M, \mathbb{R}(n))$$

have the same dimension since there are no (p,p)-classes in H^w_B. This is sufficient for (5.1). Note that multiplication by T^{-e} where $e=[K:\mathbb{R}]$ on $H^w_{ar}(M)$ maps $H^w_{ar}(M)^{\Theta=m}$ to $H^w_{ar}(M)^{\Theta=m-e}$ and thus provides some explanation for the periodicity of $\mathscr{D}$-cohomology for large n.

6. B_{ar}, part II

In this section we will investigate the functor $\mathbb{D}$ in more detail. The point to notice is that $\mathbb{D}$ and hence also the Γ-factors of a variety are already determined by a certain structure which the Hodge decomposition induces on the underlying real vector space.

Let $\underline{Filt}_{\mathbb{C}}$ be the additive category of finite dimensional real vector spaces V with a decreasing filtration $\mathrm{Fil}^r V$ such that $\mathrm{Fil}^{r_1} V=0$, $\mathrm{Fil}^{r_2} V=V$ for some r_1, r_2. Morphisms are supposed to respect the filtrations.

Let $\underline{Filt}_{\mathbb{R}}$ be the additive category of finite dimensional real vector spaces V with a decreasing filtration as above and with an involution F_∞ which respects the filtration and induces multiplication by $(-1)^*$ on the associated graded vector space $Gr^* V$. Morphisms are supposed to respect the filtration and to commute with F_∞.

For $K=\mathbb{C}$ or $\mathbb{R}$ the category $\underline{Filt}_K$ has a natural tensor product. For integers n we define objects $\mathbb{R}(n)=(2\pi i)^n\mathbb{R}$ with filtration

$$\mathrm{Fil}^r\mathbb{R}(n)=\begin{cases}\mathbb{R}(n) & \text{for } r\leqq -n\\ 0 & \text{for } r>-n\end{cases}$$

and in case $K=\mathbb{R}$ with action of F_∞ by $(-1)^n$. As usual we write

$$V(n)=V\otimes\mathbb{R}(n)$$

for the twist of an object V of $\underline{Filt}_K$. Clearly $V(0)=V$.

On $B_{\mathbb{R}}:=B_{ar,\mathbb{R}}$ a filtration is defined by

$$\mathrm{Fil}^r B_{\mathbb{R}}=\bigoplus_{n\in\mathbb{Z}}\mathrm{Fil}^r\mathbb{R}(n)\cong T^{-r}\mathbb{R}[T^{-1}].$$

Hence we have

$$(\mathrm{Fil}^r B_{\mathbb{R}})\otimes_{\mathbb{R}}\mathbb{C}=\mathrm{Fil}^r B_{ar}.$$

For an object V of $\underline{Filt}_K$ we endow $V\otimes B_{\mathbb{R}}$ with the tensor product filtration (and Frobenius if $K=\mathbb{R}$) and set:

$$\begin{aligned}\mathbb{D}(V)&=\mathrm{Fil}^0(V\otimes_{\mathbb{R}}B_{\mathbb{R}}) && \text{if } K=\mathbb{C}\\ \mathbb{D}(V)&=\mathrm{Fil}^0(V\otimes_{\mathbb{R}}B_{\mathbb{R}})^{F_\infty=\mathrm{id}} && \text{if } K=\mathbb{R}\end{aligned}\tag{6.1}$$

viewed as $\mathbb{L}$-modules with a derivation Θ over $\Theta_{\mathbb{L}}$.

We define a splitting of an object V of $\underline{Filt}_K$ to be a direct sum decomposition into subspaces V^p

$$V=\bigoplus_p V^p$$

such that

$$\mathrm{Fil}^r V=\bigoplus_{r\leqq p}V^p$$

and in case $K=\mathbb{R}$

$$F_\infty|_{V^p}=(-1)^p.$$

Splittings exist and are not unique in general. In terms of a splitting of V we have

$$\mathbb{D}(V)=\bigoplus_p V^p\otimes T^p\mathbb{L}.\tag{6.2}$$

(6.3) Proposition. *The construction $\mathbb{D}$ defines an additive functor which commutes with tensor products and maps $\mathbb{R}(n)$ to $\mathbb{L}(n)$:*

$$\mathbb{D}:\underline{Filt}_K\to\underline{Der}_K.$$

For V in $\underline{Filt}_K$ we have

$$\dim_{\mathbb{R}}V=\mathrm{rk}_{\mathbb{L}}\mathbb{D}(V).$$

The natural map

$$\xi_V:\mathbb{D}(V)\otimes_{\mathbb{L}}B_{\mathbb{R}}\to V\otimes_{\mathbb{R}}B_{\mathbb{R}}\tag{6.4}$$

induced by multiplication on $B_{\mathbb{R}}$ is bijective.

Both ξ_V and ξ_V^{-1} respect the filtrations, infinite Frobenius, derivation Θ and $\mathbb{L}$-module structure on both sides.

Proof. The tensor product of splittings of V_1 and V_2 in $\underline{Filt}_K$ is a splitting of $V_1\otimes V_2$. Using (6.2) compatibility of $\mathbb{D}$ with tensor products follows. The remaining assertions follow by elementary calculations.

For an object D of $\underline{Der}_K$ we endow $D\otimes_{\mathbb{L}} B_{\mathbb{R}}$ with the derivation $\Theta=\Theta_D\otimes\mathrm{id}+\mathrm{id}\otimes\Theta_{B_{\mathbb{R}}}$. On the real vector space

$$\mathbb{H}(D)=(D\otimes_{\mathbb{L}} B_{\mathbb{R}})^{\Theta=0}$$

we then have a filtration

$$\mathrm{Fil}^r\,\mathbb{H}(D)=(D\otimes_{\mathbb{L}}\mathrm{Fil}^r B_{\mathbb{R}})^{\Theta=0}\subset\mathbb{H}(D)$$

and in case $K=\mathbb{R}$ the infinite Frobenius $\mathrm{id}\otimes F_\infty$. If $e_1,\dots,e_n$ is an $\mathbb{L}$-basis of D we have

$$\Theta(e_j)=\sum_i a_{ij}e_i$$

and Θ is completely determined by the matrix $A=(a_{ij})$ in $M_n(\mathbb{L})$. In these terms $\mathbb{H}(D)$ is given by the space of vectors $x=(x_1,\dots,x_n)^t$ in $B_{\mathbb{R}}^n$ which satisfy the first order linear differential equation

$$\left(T\frac{d}{dT}\right)(x)+Ax=0. \tag{6.5}$$

From the general theory of such equations we know that

$$\dim_{\mathbb{R}}\mathbb{H}(D)\leqq\mathrm{rk}_{\mathbb{L}}D. \tag{6.6}$$

It is now immediate that the construction $\mathbb{H}$ defines an additive functor

$$\mathbb{H}:\underline{Der}_K\to\underline{Filt}_K$$

which maps $\mathbb{L}(n)$ to $\mathbb{R}(n)$. It follows from (6.3) that for V in $\underline{Filt}_K$ the natural map

$$\mathbb{H}(\mathbb{D}(V))\to V$$

is an isomorphism in $\underline{Filt}_K$.

Pursuing the analogy with Fontaine's theory [F1] we call an object D of $\underline{Der}_K$ admissible if

$$\dim_{\mathbb{R}}\mathbb{H}(D)=\mathrm{rk}_{\mathbb{L}}D.$$

In our case the admissible objects have in fact a very simple structure. To prove this we need an analogue of a well known lemma by Tate [Se1] § 2, Prop. 4. Recall that $\mathbb{L}=\mathbb{R}[T^{-e}]$ where $e=[K:\mathbb{R}]$ is the ramification index of K over $\mathbb{R}$.

(6.7) Lemma. *For D in $\underline{Der}_K$ the natural map of $\mathbb{R}$-vector spaces*

$$\bigoplus_{r\in\mathbb{Z}}(D^{\Theta=r}/T^{-e}D^{\Theta=r+e})\to D/T^{-e}D$$

is injective.

Proof. It is clear that for each r the natural map

$$D_r=D^{\Theta=r}/T^{-e}D^{\Theta=r+e}\to D/T^{-e}D$$

is injective. If the lemma is false there exist integers $r_1<r_2<\dots<r_n$ and elements d_{r_i} in $D^{\Theta=r_i}$ for $i=1,\dots,n$ such that the class $[d_{r_i}]$ of d_{r_i} in D_{r_i} is non-zero for each i and such that

$$\sum_{i=1}^{n} d_{r_i}=T^{-e}d \quad\text{for some } d \text{ in } D. \tag{6.8}$$

Let us assume that n is minimal with this property. By the above remark we have $n \geqq 2$. Applying Θ to (6.8) we get

$$\sum_{i=1}^{n} (r_i+e)d_{r_i} = T^{-e}\Theta(d)$$

and hence

$$\sum_{i=1}^{n-1} (r_n - r_i)d_{r_i} = T^{-e}((r_n+e)d - \Theta(d)).$$

This is a relation of the above type of shorter length $n-1 \geqq 1$. Contradiction!

Note that the proof of (6.7) is analogous to the proof of Tate's lemma.

Consider the natural map induced by the multiplication on $B_{\mathbb{R}}$:

$$\eta_D : \mathbb{H}(D) \otimes_{\mathbb{R}} B_{\mathbb{R}} \to D \otimes_{\mathbb{L}} B_{\mathbb{R}}.$$

It respects the natural filtration, the $(\mathbb{L}, \Theta)$-module structure and in case $K = \mathbb{R}$ the infinite Frobenius on both sides. In fact η_D is even strictly compatible with the filtrations:

(6.9) Corollary. *The map induced by η_D on the associated graded $\mathbb{R}$-vector spaces*

$$Gr^{\cdot}(\eta_D) : Gr^{\cdot}(\mathbb{H}(D) \otimes_{\mathbb{R}} B_{\mathbb{R}}) \to Gr^{\cdot}(D \otimes_{\mathbb{L}} B_{\mathbb{R}})$$

is injective.

Proof. There is an isomorphism

$$Gr^i \mathbb{H}(D) = \frac{(D \otimes_{\mathbb{L}} \mathrm{Fil}^i B_{\mathbb{R}})^{\Theta=0}}{(D \otimes_{\mathbb{L}} \mathrm{Fil}^{i+1} B_{\mathbb{R}})^{\Theta=0}} \xrightarrow{\sim} \frac{D^{\Theta=i}}{T^{-e} D^{\Theta=i+e}}$$

induced by mapping $d \otimes_{\mathbb{L}} T^{-i}$ to d. Since

$$Gr^j(D \otimes_{\mathbb{L}} B_{\mathbb{R}}) = D \otimes_{\mathbb{L}} (\mathrm{Fil}^j B_{\mathbb{R}} / \mathrm{Fil}^{j+1} B_{\mathbb{R}}) \xrightarrow{\sim} D / T^{-e} D$$

via $d \otimes_{\mathbb{L}} [T^{-j}] \mapsto [d]$ the corollary follows from (6.7).

Note the following consequence

(6.10) Corollary. *For D in $\underline{Der}_K$ the map η_D is injective. If it is surjective then the inverse map η_D^{-1} is also compatible with filtration, $(\mathbb{L}, \Theta)$-and Frobenius structure.*

Proof. Since $\mathbb{H}(D)$ lies in $\underline{\underline{Filt}}_K$ we have $\mathrm{Fil}^{r_1} \mathbb{H}(D) = 0$ for some r_1. Hence

$$\mathrm{Fil}^t(\mathbb{H}(D) \otimes_{\mathbb{R}} B_{\mathbb{R}}) = \sum_{\substack{r+s=t \\ r \leqq r_1}} (\mathrm{Fil}^r \mathbb{H}(D) \otimes_{\mathbb{R}} \mathrm{Fil}^s B_{\mathbb{R}})$$

$$\subset \mathbb{H}(D) \otimes_{\mathbb{R}} \mathrm{Fil}^{t-r_1} B_{\mathbb{R}} \quad \text{and thus}$$

the intersection of all $\mathrm{Fil}^t(\mathbb{H}(D) \otimes_{\mathbb{R}} B_{\mathbb{R}})$ is zero. Now (6.9) implies that η_D is injective and if it is also surjective that η_D^{-1} is compatible with the filtration.

We can now clarify the structure of admissible objects:

(6.11) Proposition. *For D in $\underline{Der}_K$ the following are equivalent:*

(1) *D is admissible*

(2) *D is isomorphic to an object of the form $\mathbb{D}(V)$ for some V in $\underline{\underline{Filt}}_K$*

(3) *D is isomorphic to an object of the form*

$$\bigoplus_{i=1}^{n} \mathbb{L}(k_i) \quad \text{with integers} \quad k_1, \ldots, k_n$$

(4) *The map*

$$\eta_D : \mathbb{H}(D) \otimes_{\mathbb{R}} B_{\mathbb{R}} \to D \otimes_{\mathbb{L}} B_{\mathbb{R}}$$

is bijective and identifies the filtrations, $(\mathbb{L}, \Theta)$*-and Frobenius structures on both sides.*

(5) *The natural map*

$$\mathbb{D}(\mathbb{H}(D)) \to D$$

induced by η_D *is an isomorphism.*

Proof. Because of (6.2) assertion (2) implies (3). It is immediate that (3) implies (4). The implications (4) $\Rightarrow$ (5) $\Rightarrow$ (2) and in view of (6.3) also (5) $\Rightarrow$ (1) are obvious. It will thus suffice to prove (1) $\Rightarrow$ (4) i.e. because of (6.10) that for admissible D the map η_D is surjective. We have to exhibit a $B_{\mathbb{R}}$-basis $v_1, \ldots, v_n$ of $D \otimes_{\mathbb{L}} B_{\mathbb{R}}$ such that

$$\Theta(v_1) = \ldots = \Theta(v_n) = 0 .$$

The equality

$$\dim_{\mathbb{R}} \mathbb{H}(D) = \mathrm{rk}_{\mathbb{L}} D$$

means that the differential equation (6.5) has a fundamental system of solutions

$$X = (x^{(1)}, \ldots, x^{(n)}) \quad \text{with} \quad x^{(i)} \quad \text{in} \quad B_{\mathbb{R}}^n$$

$$\left(T \frac{d}{dT}\right) X + AX = 0 . \tag{6.12}$$

Viewing Laurent polynomials in $B_{\mathbb{R}}$ as functions on $\mathbb{C}^*$ this implies that X is invertible at every point of $\mathbb{C}^*$. Hence $\det X$ is a Laurent polynomial without zeros or poles on $\mathbb{C}^*$ i.e.

$$\det X = \lambda T^a \quad \text{for some integer } a \text{ and } \lambda \text{ in } \mathbb{R}^* .$$

Therefore the entries of X^{-1} are again Laurent polynomials and hence X is an element of $Gl_n(B_{\mathbb{R}})$. The chosen basis $e_1, \ldots, e_n$ of D provides an identification of $D \otimes_{\mathbb{L}} B_{\mathbb{R}}$ with $B_{\mathbb{R}}^n$. Define $v_1, \ldots, v_n$ to be the $B_{\mathbb{R}}$-basis of $D \otimes_{\mathbb{L}} B_{\mathbb{R}}$ corresponding to the column vectors $x^{(1)}, \ldots, x^{(n)}$ of X. Then $\Theta(v_1) = \ldots = \Theta(v_n) = 0$.

(6.13) Remark. In terms of bases the equivalence between (1) and (3) means the following:

Given A in $M_n(\mathbb{L})$, the system on $\mathbb{C}^*$:

$$T \frac{d}{dT}(x) + Ax = 0$$

has a fundamental solution X whose entries are real Laurent polynomials if and only if there exists a matrix

$$P \text{ in } GL_n(\mathbb{L})$$

such that

$$PAP^{-1} + P\left(T \frac{d}{dT}\right) P^{-1} = -\begin{pmatrix} k_1 & & \\ & \ddots & \\ & & k_n \end{pmatrix} \quad \text{for integers } k_i .$$

This means that via $y = Px$ the above system can be transformed into

$$T \frac{d}{dT}(y) = \begin{pmatrix} k_1 & & \\ & \ddots & \\ & & k_n \end{pmatrix} y .$$

Let $\underline{Der}_K^{ad}$ denote the full subcategory of $\underline{Der}_K$ generated by the admissible objects. The first half of the following corollary is now obvious:

(6.14) Corollary. *The functors* $\mathbb{D}$ *and* $\mathbb{H}$ *provide quasi-inverse equivalences of categories:*

$$\underline{Filt}_K \underset{\mathbb{H}}{\overset{\mathbb{D}}{\rightleftarrows}} \underline{Der}_K^{ad}.$$

For D_1, D_2 *in* $\underline{Der}_K^{ad}$ *we have*

$$\mathbb{H}(D_1 \otimes_{\mathbb{L}} D_2) \cong \mathbb{H}(D_1) \otimes_{\mathbb{R}} \mathbb{H}(D_2)$$

in $\underline{Filt}_K$ *naturally.*

Proof. For D_1, D_2 in $\underline{Der}_K$ we have a natural map

$$\mathbb{H}(D_1) \otimes_{\mathbb{R}} \mathbb{H}(D_2) \to \mathbb{H}(D_1 \otimes_{\mathbb{L}} D_2).$$

For admissible objects D_1, D_2 if follows from (6.11) (3) that this is an isomorphism in the category $\underline{Filt}_K$.

Finally we make the connection with the theory of Sect. 3.

Define an additive functor $\mathscr{V}$ by composition:

$$\mathscr{V} = \mathbb{H} \circ \mathbb{D} : \underline{Hod}_K \to \underline{Der}_K^{ad} \to \underline{Filt}_K$$

There is thus an isomorphism of functors

$$\mathbb{D} \cong \mathbb{D} \circ \mathscr{V} : \underline{Hod}_K \to \underline{Der}_K^{ad}.$$

$\mathscr{V}$ commutes with twists and in case $K = \mathbb{C}$ also with tensor products. On an object H in $\underline{Hod}_K$ of type $H = H^{pp}$ or $H = H^{pq} \oplus H^{qp}$, $p < q$ the functor $\mathscr{V}$ is given by $\mathscr{V}(H) = H_{\mathbb{R}}$ with filtration:

$$\mathrm{Fil}^r H_{\mathbb{R}} = \begin{cases} H_{\mathbb{R}} & \text{for } r \leqq p \\ 0 & \text{for } r > p \end{cases} \quad \text{if } K = \mathbb{C}$$

$$\mathrm{Fil}^r H_{\mathbb{R}} = \begin{cases} H_{\mathbb{R}} & \text{for } r < p \\ H_{\mathbb{R}}^{F_\infty = (-1)^p} & \text{for } r = p \\ 0 & \text{for } r > p \end{cases} \quad \text{if } K = \mathbb{R}.$$

In case $K = \mathbb{R}$ the Frobenius F_∞ on $\mathscr{V}(H)$ is the obvious one. Note that it acts by multiplication with $(-1)^{\cdot}$ on $Gr^{\cdot}\mathscr{V}(H)$ as it must.

For H in $\underline{Hod}_K$ pure of weight w the object $\mathscr{V}(H)$ in $\underline{Filt}_K$ determines the isomorphism class of H in $\underline{Hod}_K$.

Acknowledgements. It is a pleasure for me to thank Peter Schneider for discussions and valuable comments. In particular he made the crucial suggestion to look for a "Barsotti-Tate ring at infinity". I would also like to thank the Institut d'Etudes Catalans where part of this work was done for financial support and friendly atmosphere.

References

[B] Beilinson, A.A.: Higher regulators and values of L-functions. J. Sov. Math. **30**, 2036–2070 (1985)

[E] Erdelyi, A.: Higher transcendental functions Vol. I. Bateman Manuscript project. New York Toronto London: McGraw-Hill 1953

[F1] Fontaine, J.-M.: Modules galoisiens, modules filtrés et anneaux de Barsotti-Tate. In: Journée dé Géométrie Algébrique de Rennes, Astérisque **65**, 3–80. Société Math. de France, Paris 1979

[F2] Fontaine, J.-M.: Sur certains types de représentations p-adique du groupe de Galois d'un corps local; construction d'un anneau de Barsotti-Tate. Ann. Math. **115**, 529–577 (1982)

[H] Hawking, S.W.: Zeta function regularization of Path Integrals in curved spacetime. Comm. Math. Phys. **55**, 133–148 (1977)

[J] Jannsen, U.: Deligne homology, Hodge-$\mathscr{D}$-conjecture and motives. In: Rapoport, M., Schappacher, N., Schneider, P. (eds.) Beilinson's conjectures on special values of L-functions, 305–372, (Perspectives in Math., Vol. 4) Boston New York: Academic Press 1988

[R-S] Ray, D.B., Singer, I.M.: Analytic torsion for complex manifolds. Ann. Math. **98**, 154–177 (1973)

[Sch] Schneider, P.: Introduction to the Beilinson conjectures. In: Rapoport, M., Schappacher, N., Schneider, P. (eds) Beilinson's conjectures on special values of L-functions, 1–35, (Perspectives in Math., Vol. 4) Boston New York: Academic Press 1988

[Se1] Serre, J.P.: Sur les groupes de Galois attachés aux groupes p-divisibles. In: Proceedings of a conference on local fields, Nuffic Summer school at Driebergen, 118–131, Berlin Heidelberg New York: Springer 1967

[Se2] Serre, J.P.: Facteurs locaux des fonctions zêta des variétés algébriques (définitions et conjectures). Sém. Delange-Pisot-Poitou, exp. 19, 1969/70